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Impact of SGLT2 Inhibitors on Kidney Function Among People with HIV: A US Prospective Multi-Site Study

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Abstract

Background: Sodium-glucose cotransporter 2 inhibitors (SGLT2i) are nephroprotective but are associated with early eGFR decline, also termed “eGFR dip.” Despite elevated risk of kidney disease, this phenomenon is understudied among people with HIV (PWH).

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Methods: In a 1:1 propensity score-matched cohort study using data from 9 Centers for AIDS Research Network of Integrated Clinical Systems (CNICS) sites, we compared new SGLT2i users with new users of other antihyperglycemic classes. We estimated adjusted hazard ratios (aHRs) for time to ≥10% and ≥30% eGFR decline using multivariable Cox proportional hazards models and analyzed eGFR change using multivariable linear mixed models. Additionally, longer-term eGFR trends over 24 months were visualized using locally weighted scatterplot smoothing (LOWESS) curves.

Results: Among 1554 eligible PWH, we obtained 295 matched pairs. Over 6 months, eGFR decline incidence of ≥10% and ≥30% was higher among users of SGLT2i versus other antihyperglycemic classes, (58.2% vs. 37.4%; aHR: 1.79; 95% CI: 1.40-2.28; and 17.3% vs. 9.8%; aHR: 1.69; 95% CI: 1.05-2.73; respectively) Adjusted mean eGFR change at 6 months was −2.62 mL/min/1.73m^2 for SGLT2i versus 0.05 mL/min/1.73m^2 for other classes. Long-term eGFR trends revealed an expected initial decline following SGLT2i initiation, followed by stabilization and a slower subsequent decline compared to other classes.

Conclusion: Acute eGFR dips were more common among PWH initiating SGLT2i relative to other antihyperglycemic classes, though overall declines were small, transient, and consistent with the general population. Further research is needed to explore the long-term effects of SGLT2i in PWH.

Keywords

SGLT2 inhibitors; People with HIV; Acute eGFR dip; Antihyperglycemic medications; kidney function

Introduction

In the past decade, sodium-glucose co-transporter-2 inhibitors (SGLT2i) have received attention for their cardiorenal benefits, with large clinical trials demonstrating a reduction in kidney disease progression, irrespective of diabetes status, and major cardiovascular events.[1–7] Consequently, SGLT2i are of interest for the clinical management of people with HIV (PWH), as cardiovascular disease and chronic kidney disease (CKD) are more prevalent among PWH than the general population, especially as the population of PWH ages. [8–12]

A well-recognized early effect of SGLT2i initiation is an acute, transient decline in estimated glomerular filtration rate (eGFR), commonly referred to as the eGFR “dip,” which typically ranges from 3–6 mL/min/1.73m^2 in the general population and is usually reversible, stabilizing over time.[13–16] This dip primarily reflects hemodynamic changes rather than structural kidney injury. [17] SGLT2 inhibition reduces glucose and sodium reabsorption in the proximal tubule, increasing sodium delivery to the macula densa and triggering tubuloglomerular feedback, which causes afferent arteriole vasoconstriction and lowers intraglomerular pressure.[18] While this mechanism produces an initial eGFR decrease, it may protect against glomerular hyperfiltration and slow long-term kidney function decline.[19] Clinically, SGLT2i discontinuation is generally unnecessary unless an unexplained eGFR decline exceeds 30%.[13,20–23] PWH may be particularly susceptible to acute eGFR changes due to an already elevated risk of renal complications, including CKD and acute kidney injury (AKI), driven by a combination of HIV-specific factors, such as chronic inflammation and antiretroviral therapy (ART), as well as traditional risk factors like diabetes and hypertension.[24] As a result, PWH may require closer monitoring and potential medication adjustments when initiating SGLT2i.

Despite their long-term benefits, SGLT2i are currently underutilized in PWH.[25] Barriers to treatment include potential safety concerns, particularly regarding acute eGFR declines that remain poorly characterized within this population, as well as limited insurance coverage for these medications.[26–28] Such factors may deter clinicians from prescribing SGLT2i, making it crucial to understand these declines. Lack of clinician awareness of this therapeutic class and its benefits may also contribute to under-prescribing.[29] This study characterized the incidence and magnitude of acute eGFR declines within the first 6 months and explored long-term eGFR trends among PWH initiating SGLT2i compared to other antihyperglycemic medication classes.

Methods

Study Design and Population

We conducted an observational study using data from the Centers for AIDS Research (CFAR) Network of Integrated Clinical Systems (CNICS) cohort.[30] The CNICS cohort is a dynamic, prospective, clinical cohort of PWH aged 18 and older in care at ten academic sites across the United States. This study includes data from 9 CNICS sites with the necessary data available: University of Washington, Johns Hopkins University, University of Alabama at Birmingham, University of California San Diego, University of North Carolina at Chapel Hill, University of California San Francisco, Case Western Reserve University, Fenway Health, and Vanderbilt University. Sites received local institutional review board approval for participation in CNICS.

PWH who initiated an SGLT2i or other antihyperglycemic medications from the glucagon-like peptide-1 receptor agonist (GLP-1RA), dipeptidyl peptidase-4 inhibitor (DPP-4i), or sulfonylurea classes between 2013 and 2023 were eligible for inclusion. New users were defined as having no previously recorded prescription of any medication within these four classes, with the date of treatment initiation defined as the index date. To be included, PWH needed to have had at least one eGFR measurement in the year prior to medication initiation, hereafter referred to as the baseline eGFR, and at least one eGFR measurement within 6 months of the index date, hereafter referred to as the post-index eGFR. If a patient had more than one eGFR measurement in the year prior to initiating therapy, the measurement closest to the index date was chosen as the baseline eGFR. PWH with a baseline eGFR <15 mL/min/1.73m^2, indicative of end-stage renal disease, and those who initiated two or more medications from different antihyperglycemic classes on the index date were excluded.

Outcomes Definitions and Exposure Assessment

Primary outcomes were time to first occurrence of ≥10% and ≥30% decline in eGFR mL/min/1.73m^2 within the first 6 months following medication initiation. A ≥10% decline was used to capture acute “dipping,” whereas a ≥30% decline, a commonly used marker of clinically meaningful kidney function reduction that may warrant closer monitoring or SGLT2i discontinuation, was used to assess more severe declines.

Secondary outcomes included acute eGFR change at 6 months, assessed as the absolute difference between baseline and post-index eGFR measurements, as well as the time to a ≥20% decline in eGFR, and the time to absolute decreases in eGFR of ≥5 and ≥10 mL/min/1.73m^2. Additionally, we assessed the trajectory of eGFR over 24 months to characterize longer-term trends in eGFR. Exploratory analyses evaluated potential predictors of a ≥10% eGFR decline within 6 months of medication initiation. All analyses compared users of SGLT2i with a combined group of other antihyperglycemic classes, including GLP-1RA, DPP-4 inhibitors, and sulfonylureas. eGFR was calculated using the race-free 2021 Chronic Kidney Disease Epidemiology Collaboration (CKD-EPI) equation.[31,32]

We identified medication use based on prescription records from routine HIV care visits. Treatment categories included the following: SGLT2i: empagliflozin, canagliflozin, dapagliflozin, and ertugliflozin; GLP-1RA: dulaglutide, exenatide, semaglutide, liraglutide, lixisenatide, and albiglutide; DPP-4 inhibitors: sitagliptin, saxagliptin, linagliptin, and alogliptin; and sulfonylureas: glyburide, glimepiride, and glipizide.

Statistical Analysis

Baseline characteristics were summarized using descriptive statistics. We computed means and standard deviations (SD) for continuous variables, and calculated frequencies and percentages for categorical variables. When determining follow-up time, we used both an intention-to-treat (ITT) and an on-treatment analysis approach. In the ITT approach, PWH were followed up until the occurrence of the outcome (for time-to-event outcomes), death, or the study end date, defined as 6 months after the index date, whichever occurred first. The on-treatment analysis followed a similar approach but also censored patients at medication discontinuation or switching.

To minimize confounding, we performed 1:1 propensity score (PS) matching to balance covariates between new users of SGLT2i and those initiating other antihyperglycemic classes. Propensity scores were estimated using a logistic regression model that included baseline covariates selected a priori based on their potential to act as confounders or risk factors. These covariates included: age, sex, race/ethnicity (non-Hispanic White, non-Hispanic Black, Hispanic, and Other), eGFR (mL/min/1.73 m^2), CD4 count (cells/mm^3), HIV viral load (copies/mL), diabetes status, treated hypertension, smoking history (based on the presence of a tobacco use diagnosis, and categorized as never vs. ever smokers), and current use of medications, including ART, tenofovir disoproxil fumarate (TDF), insulin, metformin, and statins. Based on CNICS operational definitions, diabetes was defined as meeting at least one of the following criteria: 1) hemoglobin A1c ≥6.5%, 2) use of a diabetes-specific medication (e.g., insulin, sulfonylureas), or 3) use of a diabetes-related medication (a medication not exclusively used for diabetes (e.g., metformin)) subsequent to an earlier diabetes diagnosis.[33] Treated hypertension required both a hypertension diagnosis and a prescription for antihypertensive medication(s).

PS matching was performed using the nearest neighbor method without replacement and a caliper of 0.1.[34] Covariate balance was assessed using the standardized mean difference (SMD), with an SMD <10% indicating good balance. After matching, body mass index (BMI) and diabetes status remained imbalanced (SMD >10%). BMI was excluded from the PS model due to missing data (1.5% missingness), as including it would have reduced the number of matched individuals and compromised study power given the relatively small sample size. To address this post-matching imbalance, multiple imputation by chained equations (10 imputations) was performed for BMI, and outcome models were adjusted for BMI and diabetes, the two covariates that remained imbalanced after matching.

Time-to-event outcomes of eGFR decline were assessed using Cox proportional hazards models in the matched cohort with a robust variance estimator, adjusting for BMI and diabetes. For these analyses, the ITT follow-up definition was used. Results were reported as adjusted hazards ratios (aHR) and 95% confidence intervals (CI). Schoenfeld residuals were used to assess the proportional hazards assumption.[35] A Kaplan-Meier analysis was performed to estimate the probabilities of ≥10% and ≥30% eGFR declines at 6 months, with Kaplan-Meier curves depicting the cumulative incidence of first observed decline over time.

For the eGFR change analysis, PWH were censored when they stopped treatment and were required to have at least one on-treatment eGFR measurement during the 6-month follow-up period. As a result, the number of PWH included in the on-treatment analyses was lower than in the ITT analyses. eGFR changes were assessed using linear mixed models with random intercepts and random slopes for time, adjusting for imbalanced covariates after matching. The on-treatment definition was used for these analyses, as ITT tends to yield more conservative estimates, potentially making SGLT2i treatment appear safer.

To visualize and estimate eGFR change trends beyond six months, we examined up to 24 months of follow-up and used locally weighted scatterplot smoothing (LOWESS) to model trajectories of eGFR for PWH initiating SGLT2i versus other antihyperglycemic classes. For contextual comparison, we also plotted eGFR trajectories derived from approximate mean changes reported in published randomized controlled trials among the general population (e.g., CREDENCE, DAPA-CKD, EMPA-REG). This nonparametric method fits smooth curves to localized data without assuming a specific functional form,[36] enabling us to capture nonlinear eGFR trajectories despite fewer observations after six months.

Additionally, to evaluate predictors of a ≥10% eGFR decline we used multivariable Poisson regression with robust variance, reporting adjusted risk ratios (aRRs) and 95% CIs. Analyses were conducted overall and stratified by treatment group (SGLT2i and other antihyperglycemic classes).

Subgroup analyses for time to ≥10% eGFR decline were performed based on age (≥60 vs. <60 years), sex (male vs. female), race/ethnicity (black vs. non-black), and baseline eGFR (≥60 vs. <60 mL/min/1.73m^2), but subgroup analyses for ≥30% decline were not conducted due to the small number of events. Effect modification was tested using an interaction term between treatment group and subgroup. A p-value of <0.05 was considered statistically significant. All analyses were performed using Stata version 17 (StataCorp, College Station, TX).

To assess the robustness of our findings, we conducted three sensitivity analyses: (i) on-treatment analysis for time-to-event outcomes of ≥10% and ≥30% eGFR decline, censoring follow-up at treatment discontinuation or switching; (ii) ITT analysis for acute eGFR changes, assuming continuous exposure throughout the follow-up, regardless of discontinuation or switching; (iii) multivariable Cox regression analysis using the full cohort without applying PS matching, adjusting for baseline covariates (as another method to control for confounding).

Results

Baseline Patient Characteristics

We identified a total of 1,554 PWH who met inclusion criteria, including 299 new users of SGLT2i and 1,255 new users of other antihyperglycemic medications (585 GLP-1RA, 200 DPP-4 inhibitors, and 470 sulfonylureas; Figure S1). After matching, 295 users of SGLT2i were matched in a 1:1 ratio to 295 users of other antihyperglycemic medications, specifically 151 GLP-1RA users, 48 DPP-4i users, and 96 sulfonylurea users.

Prior to matching, there were significant differences between the groups (Table 1). SGLT2i users were older on average (56 vs. 53 years) and more likely to be male (81% vs. 74%). Those receiving SGLT2i also had a lower mean eGFR (71 vs. 83 mL/min/1.73m^2) and included a higher proportion of PWH with moderate kidney impairment (eGFR 30–60 mL/min/1.73m^2: 30% vs. 15%). After matching, baseline characteristics were generally well balanced between the two groups, including mean age (56 vs. 57 years), baseline eGFR (71 vs. 71 mL/min/1.73m^2), and eGFR category ≥60-89 mL/min/1.73m^2 (44% vs 40%). Balance was poorer for diabetes status (69% vs. 74%) and BMI (31 vs. 33 kg/m^2).

Follow-up eGFR Measurements

During the 6-month follow-up period, the median number of eGFR measurements was similar across groups, with a median of 2 (IQR: 1–4) for SGLT2i and 2 (IQR: 1–3) for other classes. The median time to the first eGFR measurement after medication initiation was shorter for SGLT2i: 69 days (IQR: 27–125) vs. 91 days (IQR: 44–134) for other classes.

Primary Outcomes

During 6 months follow-up, a total of 263 events of eGFR decline ≥10% and 72 events of eGFR decline ≥30% occurred among 590 PWH. The Kaplan-Meier cumulative incidence of ≥10% eGFR decline was higher among those initiating SGLT2i (58.2%, 95% CI: 52.4%–64.2%) compared to other antihyperglycemic classes (37.4%, 95% CI: 34.7%–40.4%), with SGLT2i use significantly associated with an increased risk of decline (aHR 1.79 [95% CI: 1.40–2.28]) (Figures 1 and 2). For ≥30% eGFR decline, 16.9% (95% CI: 12.9–22.1) of SGLT2i users experienced this outcome, compared to 10.2% (95% CI: 7.1–14.4) of users of other antihyperglycemic classes, with SGLT2i use also associated with a higher risk compared to other classes (aHR 1.69 [95% CI: 1.05 – 2.73]) (Figures 1 and 2).

Secondary Outcomes

Change in eGFR over the 6-month follow-up period was assessed in on-treatment analysis, which included 278 on SGLT2i users and 265 on other antihyperglycemic classes. SGLT2i showed a greater absolute eGFR decline vs. other antihyperglycemic classes (−2.62 mL/min/1.73m^2 [95% CI: −7.90, 2.67] vs. +0.05 [95% CI: −3.00, 3.09]); however, the between-group difference was not statistically significant (p-interaction=0.3; Table S1). Additional secondary outcomes, a decline of ≥20% and absolute declines of ≥10 and ≥5 mL/min/1.73m^2, further showed higher acute eGFR decline risk with SGLT2i versus other classes (Figure S2).

Long-term eGFR trajectories, based on LOWESS-smoothed curves in the on-treatment population, revealed an acute decline in eGFR among SGLT2i users followed by partial recovery (Figure 3). The trajectories for the two groups crossed at approximately 14 months, after which the SGLT2i group experienced a slower rate of decline. However, only 149 SGLT2i users and 152 users of other classes had at least one eGFR measurement between 7 and 24 months, representing about 55% of the original cohort. Among these, the median number of eGFR measurements during this period was 4 (IQR 2–7 for SGLT2i; 2–9 for other classes). Exploratory multivariable analyses examining factors associated with ≥10% eGFR decline are presented in Table S2.

Subgroup Analyses

Across all subgroups, SGLT2i was consistently associated with a higher risk of a ≥10% decline in eGFR compared to other antihyperglycemic classes. No statistically significant interactions were observed across subgroups (p-interaction > 0.05), indicating no evidence of effect modification (Figure S3). Given the small subgroup sample sizes, these analyses were not powered to detect statistically significant differences and should be interpreted with caution.

Sensitivity Analyses

In the first sensitivity analysis, on-treatment results for ≥10% and ≥30% eGFR declines at 6 months were consistent with the main ITT analyses, also demonstrating higher risk among SGLT2i users versus other antihyperglycemic classes (Table S3). The second sensitivity analysis, using ITT analysis for changes in eGFR at 6 months, demonstrated smaller mean eGFR declines in new users of SGLT2i compared to the on-treatment analyses, but the overall trend of greater eGFR decline among SGLT2i users remained consistent (Table S4). This difference reflects the direct effects of SGLT2i in on-treatment analyses, while ITT likely estimate an attenuated effect by including those who discontinued or switched medications. The third sensitivity analysis, multivariable regression on the full cohort adjusted for baseline covariates, showed hazard ratios consistent with the propensity score-matched cohort (Table S5).

Discussion

In our cohort study of PWH initiating antihyperglycemic medications, we found that acute eGFR decline within 6 months was more common following SGLT2i initiation compared to other antihyperglycemic medications. However, the magnitude of the decline was, on average, small, non-clinically significant, and consistent with the eGFR dip observed in the general population.

A few studies have examined kidney function changes with SGLT2i in PWH, though data remain limited. Guiraud et al. examined 20 PWH receiving dapagliflozin or empagliflozin over a median 8.3-month follow-up and observed a small but significant median increase in serum creatinine of 17.5 μmol/L (~0.20 mg/dL), consistent with hemodynamic changes.[37] Similarly, Sise et al. studied 80 PWH receiving SGLT2i and found that 11.3% experienced AKI, a rate comparable to the 9.9% observed among PWH prescribed GLP-1s, suggesting that SGLT2i are generally safe in this population.[25] Together with our results, these findings suggest that initial eGFR changes with SGLT2i in PWH are generally modest and appear to be well tolerated.

Comparing our findings with studies of people without HIV is challenging given differences in patient populations, but such comparisons highlight both similarities and differences. In large trials and observational studies of the general population, ≥30% eGFR declines are relatively uncommon, ranging from 1 to 6%.[14,38] In our cohort, such declines were more frequent, 17% among PWH initiating SGLT2i and 10% among those on other antihyperglycemic classes. This higher incidence may reflect HIV-specific factors, including chronic inflammation, immune activation, [39–43] nephrotoxic effects of certain older ART medications (such as TDF),[44,45] or the impact of some ART agents, such as dolutegravir and cobicistat, which can inhibit tubular creatinine secretion and lead to mild increases in serum creatinine that do not represent true declines in kidney function.[46] Despite this, the magnitude of the average decline with SGLT2i was similar to that observed in HIV-negative populations. Moreover, the trajectory of eGFR over 24 months mirrored that of HIV-negative populations, with an initial decline followed by recovery and stabilization. Reassuringly, these patterns are consistent with those seen in large trials, although the smaller sample size after 6 months in our cohort limits conclusions about long-term nephroprotection.

From a clinical perspective, these findings provide reassurance that the modest eGFR dip following SGLT2i initiation in PWH is not harmful and should not be interpreted as AKI.[47] In fact, studies in the general population suggest that SGLT2i reduce the risk of AKI while providing substantial cardiorenal benefits, even among patients who experience an initial dip.[14,38,48] Nevertheless, given that PWH may have increased susceptibility to clinically significant eGFR declines (≥30%), and although current guidelines do not recommend additional renal monitoring solely for the initiation of a SGLT2i,[23] clinicians should consider individualized monitoring based on the patient's overall risk profile. Recommendations for the general population, such as withholding SGLT2i if significant acute eGFR declines are observed and avoiding dose increases of angiotensin-converting enzyme (ACE) inhibitors, angiotensin II receptor blockers (ARBs), or diuretics before starting SGLT2i, would also be prudent for PWH.[22] In addition, these considerations are especially important for PWH, as many ART medications, particularly nucleoside reverse transcriptase inhibitors (NRTIs), are partially or completely renally eliminated,[49] potentially requiring dose adjustments of ART to avoid toxicity in case of very low eGFR.

Strengths and Limitations

Strengths of our study include leveraging real-world data from CNICS, a large, prospective cohort of PWH in routine clinical care, allowing our findings to be generalizable to diverse geographic and demographic populations of PWH.

The study has several limitations. Firstly, as an observational study, residual confounding is possible despite PS matching, which only accounts for measured variables. Important covariates such as ACE inhibitors, ARBs, diuretics, and nonsteroidal anti-inflammatory drugs (NSAIDs) were not included and may have introduced bias. Pre-matching differences between groups suggest potential channeling bias, with SGLT2i more often prescribed to PWH with more advanced kidney disease. While PS matching helped address baseline imbalances, some residual differences remained. Additionally, PWH on SGLT2i were more likely to undergo earlier eGFR measurements, potentially leading to differential surveillance bias and an overestimation of eGFR decline. Furthermore, due to variable timing of eGFR measurements, including some PWH with only a single follow-up, the Kaplan–Meier curves reflect the first observed decline and may not capture early, transient, or subsequent changes. This approach mirrors real-world clinical follow-up, where the timing of laboratory assessments varies across patients. Moreover, there is some potential for exposure misclassification, but if present is likely non-differential. Finally, these findings may only be generalizable to PWH in care.

Conclusion

Given the increased risk of kidney disease in PWH, evaluating new, effective therapies like SGLT2i is essential. We found that early, small eGFR declines were more common among PWH initiating SGLT2i, consistent with class effects observed in the general population. However, clinically significant eGFR declines of ≥30% were more frequent in both SGLT2i and other antihyperglycemic users compared to what is typically seen in HIV-negative populations, suggesting that HIV-related factors may increase susceptibility. While these findings are reassuring, clinicians should remain vigilant for early eGFR declines following SGLT2i initiation and consider individualized monitoring. As CKD continues to emerge as a major comorbidity, further long-term studies are needed to assess whether SGLT2i offer similar nephroprotective benefits in PWH.

Supplementary Material

Refer to Web version on PubMed Central for supplementary material.

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References

1. Neal B, Perkovic V, Mahaffey KW, et al. Canagliflozin and Cardiovascular and Renal Events in Type 2 Diabetes. N Engl J Med. 2017;377(7):644–657. [PubMed: 28605608]
2. Wanner C, Inzucchi SE, Lachin JM, et al. Empagliflozin and Progression of Kidney Disease in Type 2 Diabetes. N Engl J Med. 2016;375(4):323–334. [PubMed: 27299675]
3. Zinman B, Wanner C, Lachin JM, et al. Empagliflozin, cardiovascular outcomes, and mortality in type 2 diabetes. N Engl J Med. 2016;13(1):17–18.
4. Herrington WG, Staplin N, Wanner C, et al. Empagliflozin in Patients with Chronic Kidney Disease. N Engl J Med. 2023;388(2):117–127. [PubMed: 36331190]
5. Wiviott SD, Raz I, Bonaca MP, et al. Dapagliflozin and Cardiovascular Outcomes in Type 2 Diabetes. N Engl J Med. 2019;380(4):347–357. [PubMed: 30415602]
6. Perkovic V, Jardine MJ, Neal B, et al. Canagliflozin and Renal Outcomes in Type 2 Diabetes and Nephropathy. N Engl J Med. 2019;380(24):2295–2306. [PubMed: 30990260]
7. Heerspink HJL, Stefánsson BV., Correa-Rotter R, et al. Dapagliflozin in Patients with Chronic Kidney Disease. N Engl J Med. 2020;383(15):1436–1446. [PubMed: 32970396]
8. Shah ASV, Stelzle D, Ken Lee K, et al. Global burden of atherosclerotic cardiovascular disease in people living with HIV systematic review and meta-analysis. Circulation. 2018;138(11):1100–1112. [PubMed: 29967196]
9. Mallipattu SK, Salem F, Wyatt CM. The changing epidemiology of HIV-related chronic kidney disease in the era of antiretroviral therapy. Kidney Int. 2014;86(2):259–265. [PubMed: 24573317]
10. Islam FM, Wu J, Jansson J, Wilson DP. Relative risk of renal disease among people living with HIV: a systematic review and meta-analysis. BMC Public Health. 2012;12(1):234. [PubMed: 22439731]
11. Kalayjian RC. Renal issues in HIV infection. Curr HIV/AIDS Rep. 2011;8(3):164–171. [PubMed: 21643783]
12. McCutcheon K, Nqebelele U, Murray L, Thomas TS, Mpanya D, Tsabedze N. Cardiac and Renal Comorbidities in Aging People Living with HIV. Circ Res. 2024;134(11):1636–1660. [PubMed: 38781295]
13. Heerspink HJL, Cherney DZI. Clinical implications of an acute dip in egfr after sglt2 inhibitor initiation. Clin J Am Soc Nephrol. 2021;16(8):1278–1280. [PubMed: 33879500]
14. Adamson C, Docherty KF, Heerspink HJL, et al. Initial Decline (Dip) in Estimated Glomerular Filtration Rate After Initiation of Dapagliflozin in Patients With Heart Failure and Reduced Ejection Fraction: Insights From DAPA-HF. Circulation. 2022;146(6):438–449. [PubMed: 35442064]
15. Kraus BJ, Weir MR, Bakris GL, et al. Characterization and implications of the initial estimated glomerular filtration rate ‘dip’ upon sodium-glucose cotransporter-2 inhibition with empagliflozin in the EMPA-REG OUTCOME trial. Kidney Int. 2021;99(3):750–762. [PubMed: 33181154]
16. Oshima M, Jardine MJ, Agarwal R, et al. Insights from CREDENCE trial indicate an acute drop in estimated glomerular filtration rate during treatment with canagliflozin with implications for clinical practice. Kidney Int. 2021;99(4):999–1009. [PubMed: 33316282]
17. Yau K, Cherney DZI, van Raalte DH, Wever BE. Kidney protective mechanisms of SGLT2 inhibitors: evidence for a hemodynamic effect. Kidney Int. 2024;105(6):1168–1172. [PubMed: 38777402]
18. Vallon V. State-of-the-Art-Review: Mechanisms of Action of SGLT2 Inhibitors and Clinical Implications. Am J Hypertens. 2024;(July):841–852. [PubMed: 39017631]
19. Sen T, Heerspink HJL. A kidney perspective on the mechanism of action of sodium glucose co-transporter 2 inhibitors. Cell Metab. 2021;33(4):732–739. [PubMed: 33691091]
20. van Bommel EJM, Lytvyn Y, Perkins BA, et al. Renal hemodynamic effects of sodium-glucose cotransporter 2 inhibitors in hyperfiltering people with type 1 diabetes and people with type 2 diabetes and normal kidney function. Kidney Int. 2020;97(4):631–635. [PubMed: 32200854]
21. Umanath K, Testani JM, Lewis JB. “Dip” in eGFR: Stay the Course With SGLT-2 Inhibition. Circulation. 2022;146(6):463–465. [PubMed: 35939546]

22. Meraz-Munoz AY, Weinstein J, Wald R. eGFR Decline after SGLT2 Inhibitor Initiation: The Tortoise and the Hare Reimagined. Kidney360. 2021;2(6):1042–1047. [PubMed: 35373074]

23. KDIGO. KDIGO 2024 Clinical Practice Guideline for the Evaluation and Management of Chronic Kidney Disease. Kidney Int. 2024;105(4).

24. Heron JE, Bagnis CI, Gracey DM. Contemporary issues and new challenges in chronic kidney disease amongst people living with HIV. AIDS Res Ther. 2020;17(1):1–13. [PubMed: 31907064]

25. Sise ME, Katz-Agranov N, Strohbehn IA, et al. Use and Side Effects of Sodium Glucose Transporter 2 Inhibitors among U.S. People with HIV with Clinical Indications. JAIDS J Acquir Immune Defic Syndr. 2023;Publish Ah.

26. Yi TW, Hara DVO, Smyth B, Jardine MJ, Levin A, Morton RL. Identifying Barriers and Facilitators for Increasing Uptake of Sodium-Inhibitors in British Columbia , Canada , using the Consolidated Framework for Implementation Research. 2024;2.

27. Zhao JZ, Weinhandl ED, Carlson AM, Peter WL St. Disparities in SGLT2 Inhibitor or Glucagon-Like Peptide 1 Receptor Agonist Initiation Among Medicare-Insured Adults With CKD in the United States. Kidney Med. 2023;5(1).

28. Aggarwal R, Vaduganathan M, Chiu N, Bhatt DL. Out-of-Pocket Costs for SGLT-2 (Sodium-Glucose Transport Protein-2) Inhibitors in the United States. Circ Hear Fail. 2022;15(3):287–289.

29. Koh SWC, Lai MY, Leong CK, Lam J, Chew HSJ, Ngoh CLY. Primary Care Physicians' Perspective on SGLT2 Inhibitors for Chronic Kidney Disease. Kidney Med. 2025;7(6):101002. [PubMed: 40510618]

30. Kitahata MM, Rodriguez B, Haubrich R, et al. Cohort profile: The centers for AIDS research network of integrated clinical systems. Int J Epidemiol. 2008;37(5):948–955. [PubMed: 18263650]

31. Miller WG, Kaufman HW, Levey AS, et al. National Kidney Foundation Laboratory Engagement Working Group Recommendations for Implementing the CKD-EPI 2021 Race-Free Equations for Estimated Glomerular Filtration Rate: Practical Guidance for Clinical Laboratories. Clin Chem. 2022;68(4):511–520. [PubMed: 34918062]

32. Muiru AN, Madden E, Scherzer R, et al. Effect of Adopting the New Race-Free 2021 Chronic Kidney Disease Epidemiology Collaboration Estimated Glomerular Filtration Rate Creatinine Equation on Racial Differences in Kidney Disease Progression among People with Human Immunodeficiency Virus: An Obs. Clin Infect Dis. 2023;76(3):461–468. [PubMed: 36069064]

33. Crane HM, Kadane JB, Crane PK, Kitahata MM. Diabetes Case Identification Methods Applied to Electronic Medical Record Systems: Their Use in HIV-Infected Patients. Curr HIV Res. 2006;4(1):97–106. [PubMed: 16454715]

34. Austin PC. A comparison of 12 algorithms for matching on the propensity score. Stat Med. 2014;33(6):1057–1069. [PubMed: 24123228]

35. Schoenfeld D. Partial residuals for the proportional hazards regression model. Biometrika. 1982;69(1):239–241.

36. Cleveland WS. Robust Locally and Smoothing Weighted Regression Scatterplots. J Am Stat Assoc. 1979;74(368):829–836.

37. Guiraud V, Sauce D, Bittar R, et al. Clinical, biological, metabolic, and immune changes associated with the use of sodium-glucose cotransporter 2 inhibitors in people living with HIV. Infect Dis now. 2025;55(2):105040. [PubMed: 39961547]

38. Xie Y, Bowe B, Gibson AK, McGill JB, Maddukuri G, Al-Aly Z. Clinical implications of estimated glomerular filtration rate dip following sodium-glucose cotransporter-2 inhibitor initiation on cardiovascular and kidney outcomes. J Am Heart Assoc. 2021;10(11).

39. Rønsholt FF, Ullum Henrik; Katzenstein Terese L.; Gerstoft Jan; Ostrowski SR. Persistent Inflammation and Endothelial Activation in HIV-1 Infected Patients after 12 Years of Antiretroviral Therapy. PLoS One. 2013;8(6):1–5.

40. Hileman CO, Funderburg NT. Inflammation, Immune Activation, and Antiretroviral Therapy in HIV. Curr HIV/AIDS Rep. 2017;14(3):93–100. [PubMed: 28434169]

41. Neuhaus J, Jacobs DR, Baker JV., et al. Markers of inflammation, coagulation, and renal function are elevated in adults with HIV infection. J Infect Dis. 2010;201(12):1788–1795. [PubMed: 20446848]

42. Abraham AG, Darilay A, McKay H, et al. Kidney Dysfunction and Markers of Inflammation in the Multicenter AIDS Cohort Study. J Infect Dis. 2015;212(7):1100–1110. [PubMed: 25762788]
43. Choshi J, Hanser S, Mabhida SE, et al. A systematic review assessing the association of inflammatory markers with kidney dysfunction in people living with HIV on highly active antiretroviral therapy. BMC Infect Dis. 2024;24(1):1–9. [PubMed: 38166696]
44. Cooper RD, Wiebe N, Smith N, Keiser P, Naicker S, Tonelli M. Systematic review and meta-analysis: Renal safety of tenofovir disoproxil fumarate in hiv-infected patients. Clin Infect Dis. 2010;51(5):496–505. [PubMed: 20673002]
45. Kalayjian RC, Lau B, Mechekano RN, et al. Risk factors for chronic kidney disease in a large cohort of HIV-1 infected individuals initiating antiretroviral therapy in routine care. AIDS. 2012;26(15):1907–1915. [PubMed: 22824630]
46. Casado JL, Monsalvo M, Vizcarra P, Fontecha M, Serrano-Villar S, Moreno S. Evaluation of kidney function in HIV-infected patients receiving an antiretroviral regimen containing one or two inhibitors of the tubular secretion of creatinine. HIV Med. 2019;20(10):648–656. [PubMed: 31321875]
47. Zhao M, Sun S, Huang Z, Wang T, Tang H. Network meta-analysis of novel glucose-lowering drugs on risk of acute kidney injury. Clin J Am Soc Nephrol. 2021;16(1):70–78.
48. Jongs N, Chertow GM, Greene T, et al. Correlates and Consequences of an Acute Change in eGFR in Response to the SGLT2 Inhibitor Dapagliflozin in Patients with CKD. J Am Soc Nephrol. 2022;33(11):2094–2107. [PubMed: 35977807]
49. Holec AD, Mandal S, Prathipati PK, Destache CJ. Nucleotide Reverse Transcriptase Inhibitors: A Thorough Review, Present Status and Future Perspective as HIV Therapeutics. Curr HIV Res. 2017;15(6):100–106.

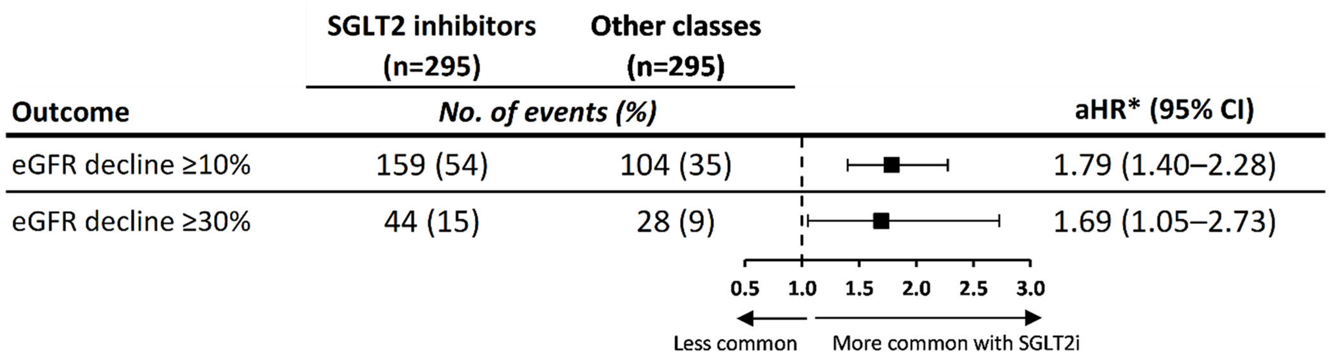

Figure 1. Adjusted hazard ratios (aHRs) for ≥10% and ≥30% estimated glomerular filtration rate (eGFR) decline within 6 months of sodium–glucose cotransporter-2 inhibitor (SGLT2i) versus other antihyperglycemic class initiation among people with HIV (PWH) in a propensity score–matched cohort (N=590).

*Models were adjusted for diabetes status and baseline BMI due to residual imbalance after matching (standardized mean difference >10%).

Abbreviations: SGLT2i, sodium–glucose cotransporter-2 inhibitor; eGFR, estimated glomerular filtration rate; aHR, adjusted hazard ratio

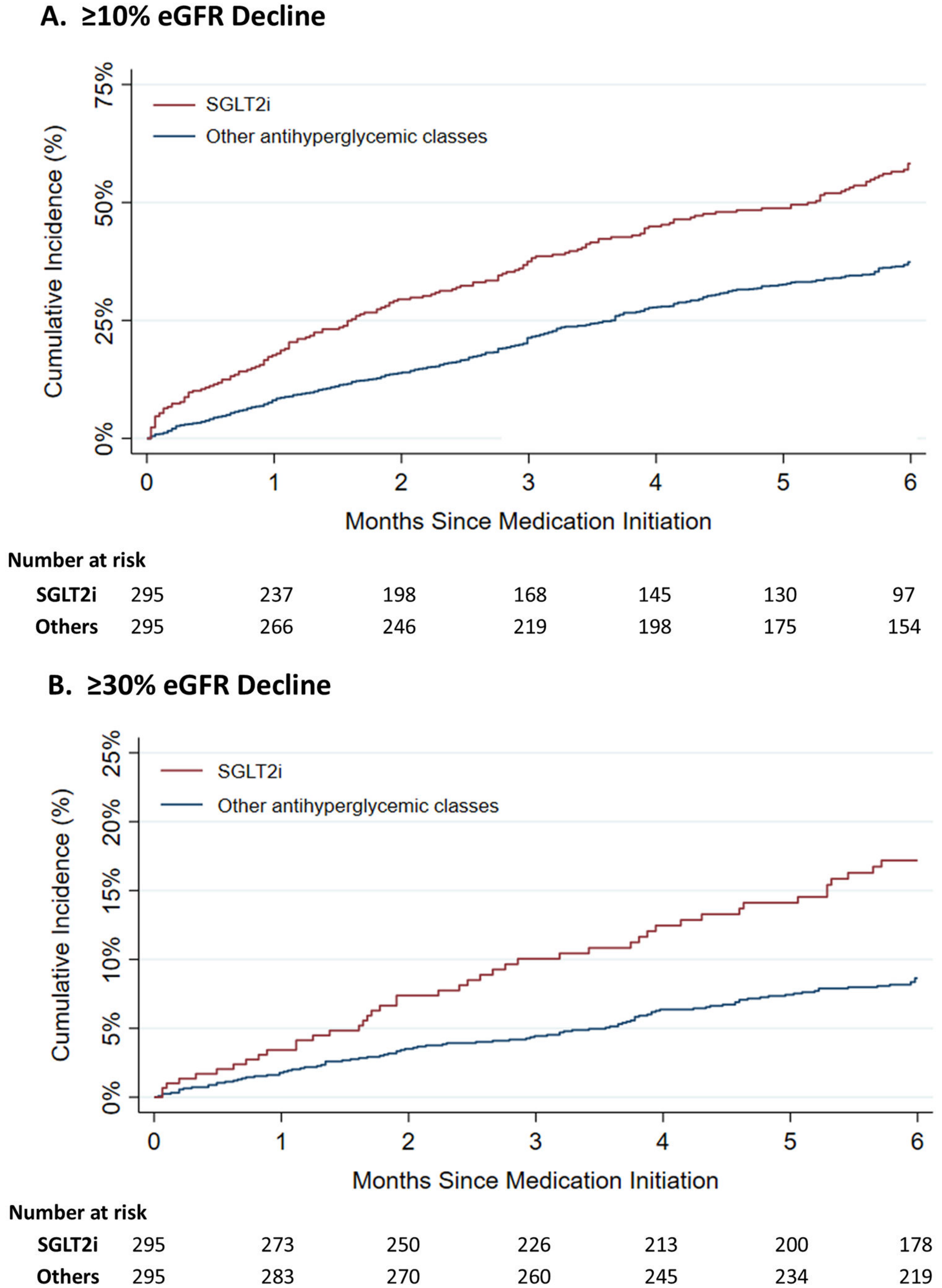

Figure 2. Kaplan-Meier curves for cumulative incidences of ≥10% (Panel A) and ≥30% (Panel B) eGFR decline within 6 Months of SGLT2 Inhibitors vs. other antihyperglycemic classes initiation in propensity score-matched cohort of PWH (N=590).

Abbreviations: SGLT2i, sodium–glucose cotransporter-2 inhibitor; eGFR, estimated glomerular filtration rate

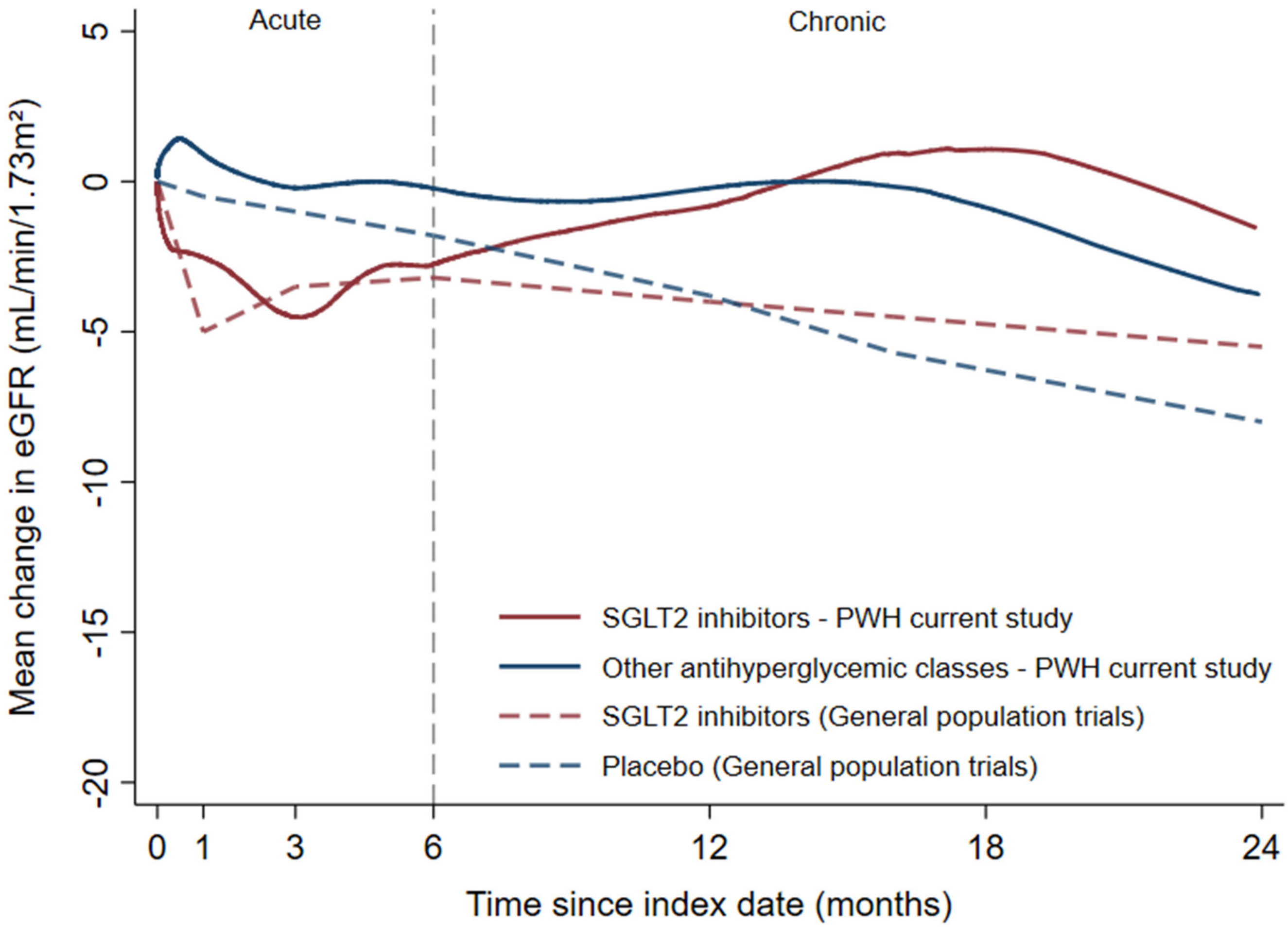

Figure 3. Change in eGFR over time following initiation of SGLT2i versus other antihyperglycemic classes among PWH (on-treatment, N = 543), with reference to general population clinical trial data.

LOWESS-smoothed eGFR trajectories over 24 months for PWH initiating SGLT2i (red line) or other antihyperglycemic classes (blue line) are overlaid with dashed lines representing approximate mean changes in eGFR reported in randomized controlled trials (RCTs) among the general population (e.g. CREDENCE, DAPA-CKD, and EMPA-REG). The eGFR trajectory in PWH follows a similar pattern to that observed in large, RCTs of SGLT2i from the general population, with an initial decline during the acute phase (0–6 months) after SGLT2i initiation, followed by partial recovery and stabilization with a crossover around 12–14 months, after which SGLT2i users exhibit a slower rate of eGFR decline compared to controls. LOWESS plots illustrate the trend in eGFR changes over time and are for descriptive purposes only.

Abbreviations: SGLT2i, sodium–glucose cotransporter-2 inhibitor; eGFR, estimated glomerular filtration rate; PWH, people with HIV; RCT, randomized controlled trial; LOWESS, locally weighted scatterplot smoothing.

Table 1.

Baseline Characteristics of People with HIV (PWH) Initiating SGLT2 Inhibitors vs. Other Antihyperglycemic Classes, Before and After Propensity Score (PS) Matching

	Before PS matching		After PS matching		
Characteristic	**SGLT2 Inhibitors (n, %)**	**Other classes (n, %)**	**SGLT2 Inhibitors (n, %)**	**Other classes [a] (n, %)**	**SMD [b] (%)**
N	299	1255	295	295	
Age, years (Mean, SD)	56 (10)	53 (10)	56 (10)	57 (10)	5.5
Sex					
Male	241 (81)	932 (74)	237 (80)	245 (83)	7.0
Female	58 (19)	323 (26)	58 (20)	50 (17)	
Race					
Non-Hispanic White	111 (37)	431 (34)	111 (38)	106 (36)	4.4
Non-Hispanic Black	147 (49)	575 (46)	143 (49)	147 (50)	
Hispanic	27 (9)	194 (16)	27 (9)	29 (10)	
Other	14 (5)	55 (4)	14 (5)	13 (4)	
eGFR, mL/min/1.73m^2 (Mean, SD)	69 (23)	83 (23)	71 (23)	71 (23)	1.7
eGFR categories					
≥ 90	70 (23)	537 (43)	70 (24)	71 (24)	8.3
≥ 60 to <90	129 (43)	500 (40)	129 (44)	119 (40)	
≥ 30 to <60	91 (30)	197 (16)	87 (30)	97 (33)	
≥ 15 to <30	9 (3)	21 (2)	9 (3)	8 (3)	
Diabetes	203 (68)	1034 (82)	203 (69)	218 (74)	11.3
Treated Hypertension	227 (76)	845 (67)	223 (76)	230 (78)	5.6
CD4 Count, cells/mL (Mean, SD)	676 (375)	724 (386)	675 (375)	665 (347)	2.7
VL <200, copies/mL	273 (91)	1157 (92)	271 (92)	274 (93)	3.8
BMI, kg/m^2 (Mean, SD)	31 (8)	34 (8)	31 (8)	33 (7)	24.0
Concomitant Baseline Medications					
ART	291 (97)	1217 (97)	287 (97)	286 (97)	2.0
TDF	44 (15)	302 (24)	44 (15)	42 (14)	1.9
DTG	119 (40)	430 (34)	117 (40)	108 (37)	6.3
RPV	28 (9)	116 (12)	28 (10)	30 (10)	2.2
COBI	29 (10)	152 (12)	29 (10)	37 (13)	8.6
Insulin	63 (21)	235 (19)	63 (21)	67 (23)	3.3
Metformin	115 (38)	669 (53)	112 (38)	124 (42)	8.3
Statins	206 (69)	688 (55)	202 (69)	204 (69)	1.5
Ever Smoker	141 (47)	451 (36)	138 (47)	130 (44)	5.5

Note:

[a] Others group consists of GLP-1 RA (n=151), DPP-4 inhibitors (n=48), and sulfonylureas (n=96).

[b] SMD <10% indicates good balance between groups.

Abbreviations: SMD: Standardized mean difference, SGLT2i: Sodium-glucose cotransporter-2 inhibitor, eGFR: Estimated glomerular filtration rate, VL: Viral load, BMI: Body mass index, ART: Antiretroviral therapy, TDF: Tenofovir disoproxil fumarate, DTG: Dolutegravir; RPV: Rilpivirine; COBI: Cobicistat.